

MATH 375: Probability Theory

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Lecture 3

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1 Lecture 3: Events and Counting Probabilities

1.1 Reading event notation

Let E and F be events. Then:

$E \cup F$	E occurs, F occurs, or both occur,
$E \cap F$	both E and F occur,
E^c	E does not occur.

Two useful probability rules are

$$P(E^c) = 1 - P(E)$$

and

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

The intersection is subtracted because it was counted once in $P(E)$ and again in $P(F)$.

We can also split F into two nonoverlapping pieces:

$$F = (F \cap E) \cup (F \cap E^c).$$

Consequently,

$$P(F) = P(F \cap E) + P(F \cap E^c).$$

1.2 When one event is contained in another

We write $E \subseteq F$ when every outcome in E is also in F . For example, when rolling a die, the event “roll a 6” is contained in the event “roll an even number.”

If $E \subseteq F$, then $E \cap F = E$. Using the previous rule,

$$\begin{aligned} P(F) &= P(F \cap E) + P(F \cap E^c) \\ &= P(E) + P(F \cap E^c) \\ &\geq P(E). \end{aligned}$$

Therefore,

$$E \subseteq F \implies P(E) \leq P(F).$$

1.3 Example: liking neither book

Jane likes the first book with probability 0.5, the second with probability 0.4, and both with probability 0.3.

Let A mean “likes the first book” and B mean “likes the second book.” The event “likes neither” is $(A \cup B)^c$. First find

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.4 - 0.3 = 0.6.$$

Then use the complement:

$$P((A \cup B)^c) = 1 - 0.6 = 0.4.$$

1.4 Equally likely outcomes

If every outcome in a finite sample space S is equally likely, then for an event E ,

$$P(E) = \frac{|E|}{|S|}.$$

Example: two dice. Two distinguishable fair dice have $6 \cdot 6 = 36$ ordered outcomes. The outcomes whose sum is 6 are

$$(1, 5), (2, 4), (3, 3), (4, 2), (5, 1).$$

There are five favorable outcomes, so

$$P(\text{sum is 6}) = \frac{5}{36}.$$

1.5 Example: selecting a committee

Select five people uniformly from a group of six men and nine women. What is the probability of selecting exactly three men and two women?

There are $\binom{15}{5}$ possible committees. For a favorable committee, choose three of the six men and two of the nine women:

$$|E| = \binom{6}{3} \binom{9}{2}.$$

Thus,

$$P(E) = \frac{\binom{6}{3} \binom{9}{2}}{\binom{15}{5}} = \frac{240}{1001}.$$

1.6 Example: a five-card straight

A five-card hand is a **straight** if its ranks are consecutive and its cards are not all of the same suit. Aces may be low ($A, 2, 3, 4, 5$) or high ($10, J, Q, K, A$), giving ten possible rank sequences.

For each sequence, there are 4^5 choices of suits. Exclude the four choices where all five cards have the same suit. Therefore,

$$|E| = 10(4^5 - 4).$$

Since there are $\binom{52}{5}$ five-card hands,

$$P(\text{straight}) = \frac{10(4^5 - 4)}{\binom{52}{5}}.$$

1.7 Example: bridge hands

In bridge, four labeled players each receive 13 cards.

One specified player receives all 13 spades. There are $\binom{52}{13}$ possible hands for that player, and only one contains all the spades. Hence

$$P(\text{player 1 gets all spades}) = \frac{1}{\binom{52}{13}}.$$

Any one player receives all 13 spades. The four possible recipients give mutually exclusive events, so

$$P(\text{some player gets all spades}) = \frac{4}{\binom{52}{13}}.$$

Each player receives exactly one ace. The number of ways to deal 52 cards to four labeled players is

$$|S| = \frac{52!}{(13!)^4}.$$

First assign the four distinct aces, one to each player, in $4!$ ways. Then divide the 48 non-ace cards into four groups of 12:

$$|E| = 4! \frac{48!}{(12!)^4}.$$

Therefore,

$$P(\text{one ace per player}) = \frac{4! 48! / (12!)^4}{52! / (13!)^4}.$$